

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Implemented backend security controls, **AWS KMS** encryption for PII/clinical data, 90-day retention, auth hardening, and rate limiting, and remediated **20+ VAPT findings**.
- Served as primary technical POC for the eTriage backend, ran knowledge transfer onboarding a second developer to on-call readiness in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Integrated LLM-powered context-aware replies into a chat product and shipped a production LLM auto-commenting feature (**4-bit quantized Llama-7B via llama.cpp**) with prompt iteration, async queued generation, and content-safety filtering.
- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with validation and clear contracts, and migrated core repositories from JavaScript to TypeScript.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN gym-management platform with Stripe + manual-cash hybrid payments, cron automation, and multi-branch support; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic service that turns open-ended queries into structured entity tables via retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided RAG pipeline (sub-question decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on 4,905 questions.

Refusal Decay – LLM Safety / Mechanistic Interpretability (GitHub) PyTorch, Llama-3.1-8B, AdvBench

- Ran a mechanistic-interpretability study of Llama-3.1-8B-Instruct in PyTorch, extracting the residual-stream refusal direction via difference-in-means (refusal **0.92** → **0.32** at $k=3$) with causal activation patching and significance testing.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, C++, TypeScript, JavaScript, SQL, Java, Bash, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: FastAPI, Node.js, React, Docker, Git, GitHub Actions, pytest, REST APIs

ML & AI: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, JAX, NLP, deep learning, computer vision

LLM & Retrieval: RAG, LLM APIs (OpenAI / Groq / Ollama), prompt engineering, RRF, cross-encoder reranking, MMR

Cloud / Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.